

ML-POWEREDANALYTICS · TEAMRETAINXAI · 2026

Customer Churn Prediction & Agentic Retention Strategy

Sector: Telecommunications — Customer Intelligence

TEAM MEMBERS	ACADEMIC DETAILS	SUBMISSION
Vanshika Yadav — Data Analysis & Model Lead Ronit Singh — Engineering Workflow Lead Riya Garg — EDA, Visualization & Deployment Lead Sankalp — Testing, Validation & GUI Lead	Faculty Advisor Prof. Bipul Shahi Course Gen-AI Group Team RetainX AI	SUBMISSION March 2026 Team RetainX AI

Customer Churn Intelligence System — Powered by XGBoost + Streamlit Deployment

SECTION 01

Project Overview

This project presents the complete design and implementation of an AI-driven customer analytics system that predicts telecom customer churn and evolves into an agentic AI-powered retention strategist. The system is structured across two milestones.

Milestone 1 — ML-Based Churn Prediction

- Process historical customer behavior data
- Identify churn patterns and predict churn probability
- Detect high-risk customers with precision
- Provide business-aligned insights for retention teams

Milestone 2 — Agentic Retention Strategy Assistant (Planned)

- Reasoning about churn probability using LLM agents
- Retrieving best retention practices using RAG (Retrieval-Augmented Generation)
- Planning structured intervention strategies autonomously
- Generating automated retention reports for decision makers

SECTION 02

Problem Statement

Customer churn is one of the most critical challenges in subscription-based industries such as telecom. Acquiring a new customer costs significantly more than retaining an existing one.

Business Area	Impact of Churn
Revenue	Loss of recurring subscription income
Customer Acquisition Cost	Higher marketing spend to replace lost users
Operational Efficiency	Wasted onboarding resources
Competitive Position	Reduction in market share

This project focuses on **early churn detection**, **probability-based risk assessment**, and **data-driven retention planning**.

SECTION 03

Dataset Information

Attribute	Value
Dataset Name	Telco Customer Churn (IBM Sample Data)
Total Records	7,043
Total Features	21 (20 inputs + 1 target variable)
Target Variable	Churn (Yes / No)
Platform	Kaggle — 2.78M views, 500K+ downloads
Final Dataset Size	7,032 (after cleaning)

Raw Dataset Preview

customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport
7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	Yes	No	No
5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes	No
3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No	No
7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	No	Yes	Yes
9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No	No
9305-CDSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	No	No	Yes	No
1452-KIOVK	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	Yes	No	No
6713-OKOMC	Female	0	No	No	10	No	No phone service	DSL	Yes	No	No	No
7892-POOKP	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	No	Yes	Yes
6388-TABGU	Male	0	No	Yes	62	Yes	No	DSL	Yes	Yes	No	No
9763-GRSKD	Male	0	Yes	Yes	13	Yes	No	DSL	Yes	No	No	No
7469-LKBCI	Male	0	No	No	16	Yes	No	No	No internet service	No internet service	No internet service	No internet service
8091-TTVAX	Male	0	Yes	No	58	Yes	Yes	Fiber optic	No	No	Yes	No
0280-XJGEX	Male	0	No	No	49	Yes	Yes	Fiber optic	No	Yes	Yes	No
5129-JLPIS	Male	0	No	No	25	Yes	No	Fiber optic	Yes	No	Yes	Yes
3655-SNQYZ	Female	0	Yes	Yes	69	Yes	Yes	Fiber optic	Yes	Yes	Yes	Yes
8191-XWSZG	Female	0	No	No	52	Yes	No	No	No internet service	No internet service	No internet service	No internet service
9959-WOFKT	Male	0	No	Yes	71	Yes	Yes	Fiber optic	Yes	No	Yes	No
4190-MFLUW	Female	0	Yes	Yes	10	Yes	No	DSL	No	No	Yes	Yes
4183-MYFRB	Female	0	No	No	21	Yes	No	Fiber optic	No	Yes	Yes	No
8779-QRDMV	Male	1	No	No	1	No	No phone service	DSL	No	No	Yes	No

Figure 1: Raw dataset snapshot — Telco Customer Churn (7,043 records, 21 columns)

Feature Categories

Category	Features
Demographics	Gender, SeniorCitizen, Partner, Dependents
Services	PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
Account Info	Tenure, Contract, PaperlessBilling, PaymentMethod
Billing	MonthlyCharges, TotalCharges

SECTION 04

Exploratory Data Analysis

EDA was conducted using **Pandas**, **Seaborn**, **Matplotlib**, **Plotly**, and **Missingno** across 7,043 rows and 21 mixed-type columns.

Missing Value Analysis

Initial missing value visualization using Missingno matrix showed no visible missing data. However, upon deeper inspection, 11 missing values were found in TotalCharges (customers with tenure = 0). These were removed. Final dataset size: **7,032 records**.

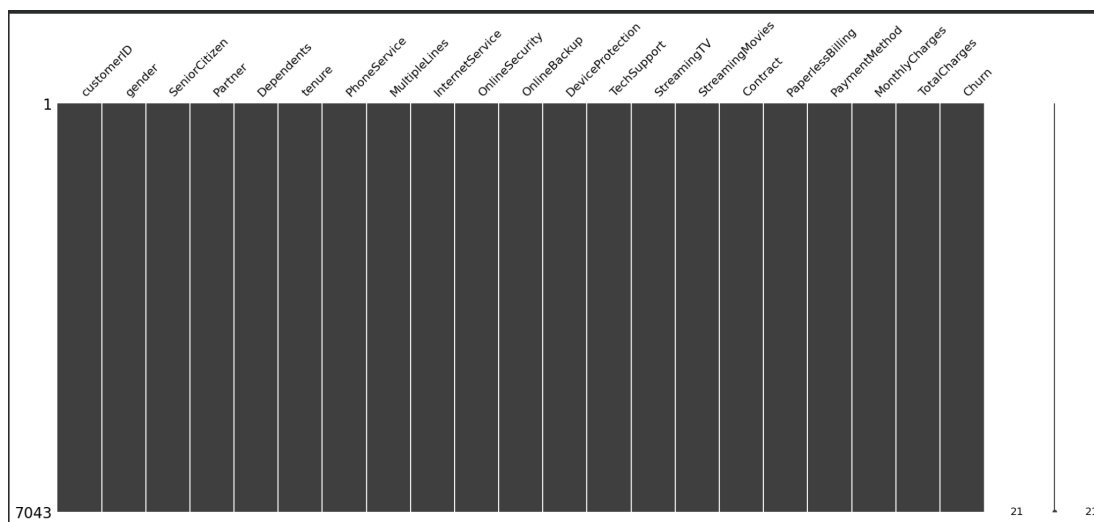


Figure 2: Missingno matrix — no visible gaps, but 11 hidden blanks found in TotalCharges

1. Contract Type

Customers with month-to-month contracts have the highest churn rate. Long-term contracts (1-year and 2-year) significantly reduce churn.

Customer contract distribution

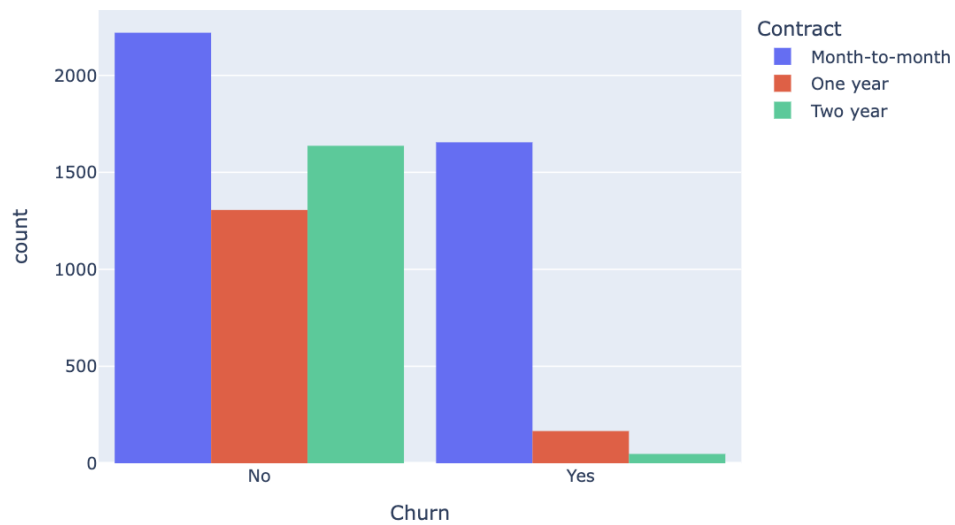


Figure 3: Customer contract distribution vs. Churn — month-to-month dominates churn

2. Payment Method

Customers using electronic checks show higher churn. Automatic bank transfer and credit card users show lower churn.

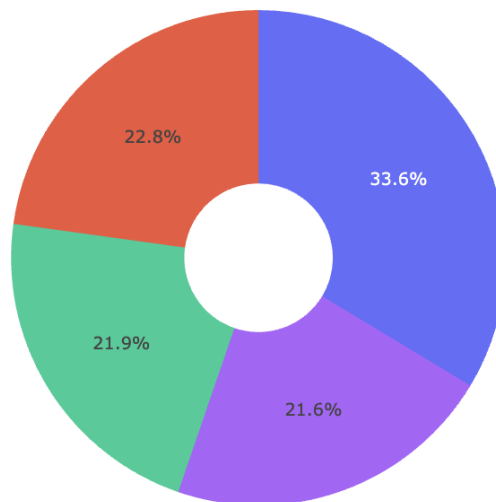


Figure 4: Payment method distribution — electronic check accounts for 33.6% of churned customers

3. Tenure

Low-tenure customers are more likely to churn. Long-tenure customers demonstrate strong retention.

Tenure vs Churn

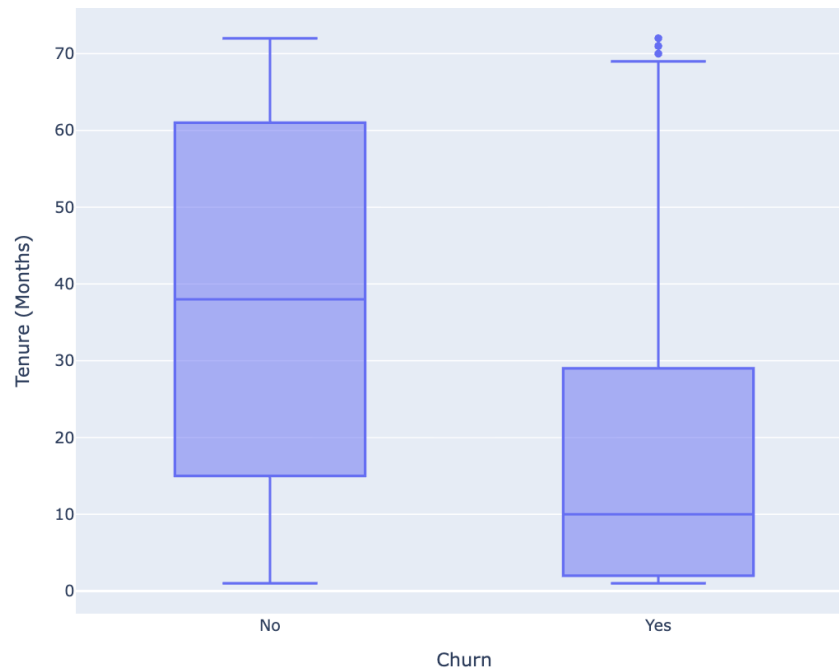


Figure 5: Tenure vs. Churn — churned customers have a median tenure far below retained customers

4. Services — Phone Service

A very small fraction of customers don't have phone service, and out of that group, 1/3 of customers are more likely to churn.



Churn distribution w.r.t. Phone Service

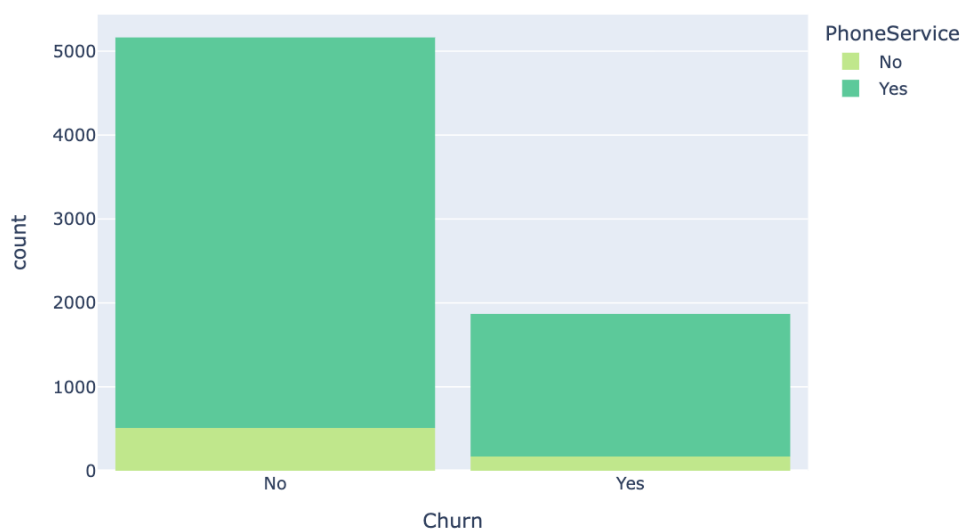


Figure 6: Churn distribution w.r.t. Phone Service

4. Services — Tech Support

Customers with no TechSupport are most likely to migrate to another service provider.

Churn distribution w.r.t. TechSupport

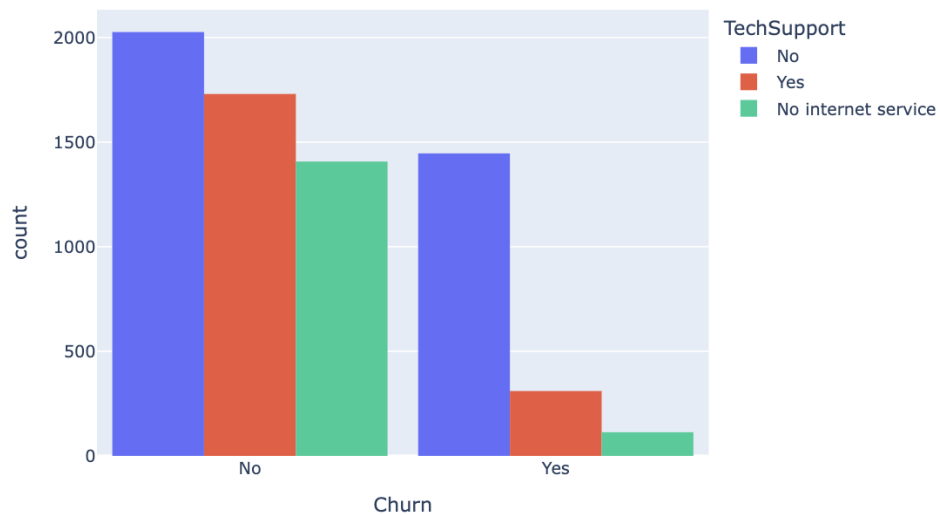


Figure 7: Churn distribution w.r.t. TechSupport — no-support customers dominate churn

4. Services — Online Security

Most customers churn in the absence of online security services.



Figure 8: Churn distribution w.r.t. Online Security

5. Monthly Charges

Higher monthly charges increase churn probability. Churned customers are concentrated in the \$60–\$100 range.

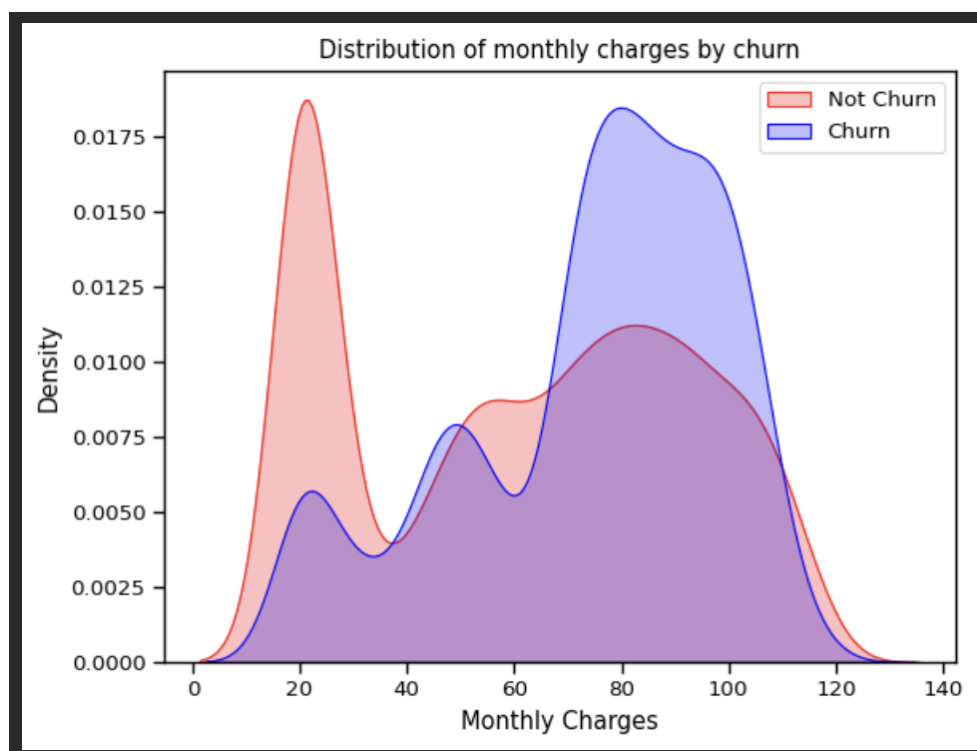


Figure 9: Distribution of monthly charges by churn — churn peaks in the high-charge segment

6. Total Charges

Lower total spending customers churn more frequently, reflecting early-stage customer vulnerability.

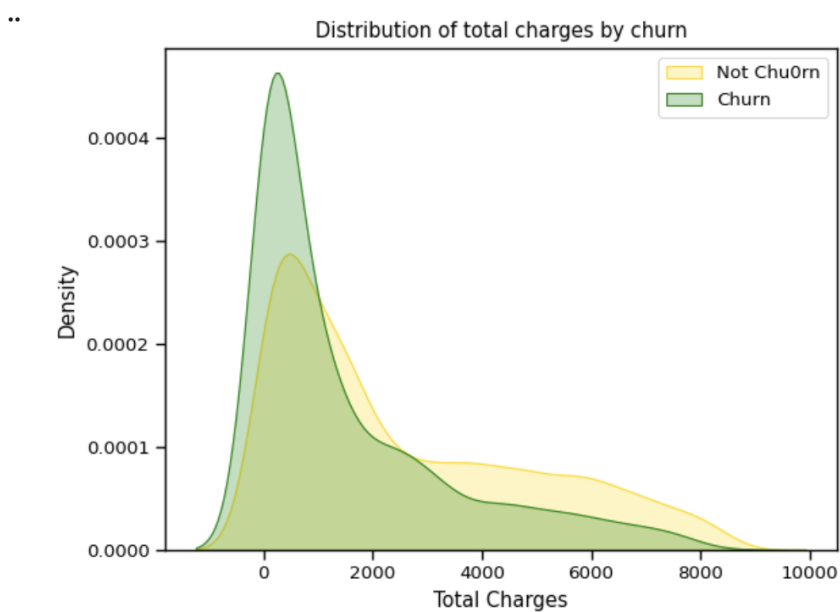


Figure 10: Distribution of total charges by churn — churned customers cluster at low total spend

7. Gender

Minimal impact on churn. Both male and female customers exhibit similar churn and retention rates — gender is not a strong predictor.

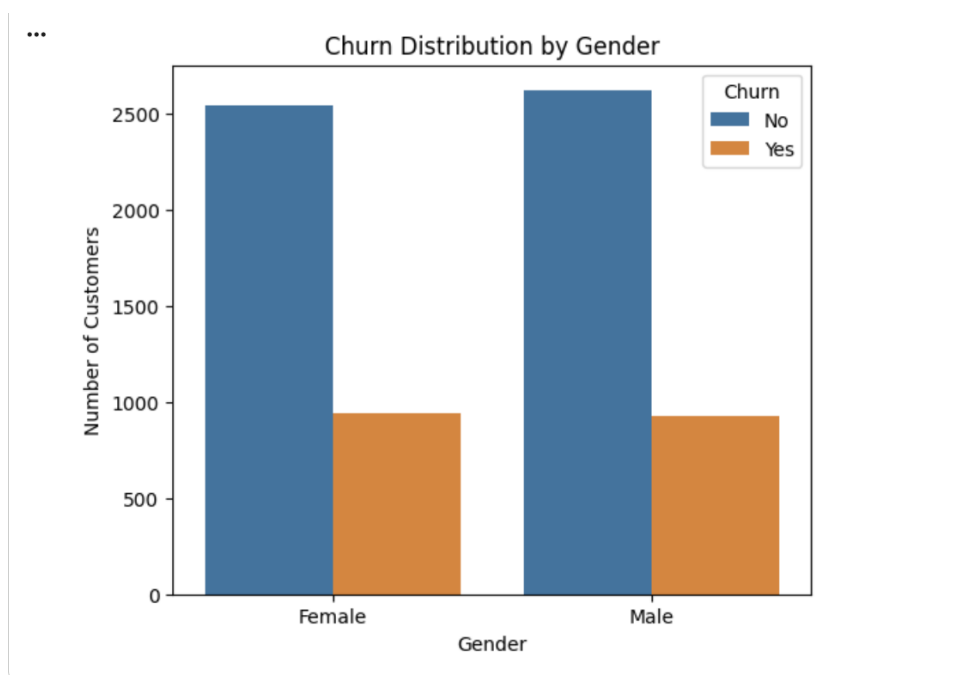


Figure 11: Churn distribution by Gender — nearly identical across both groups

Consolidated Insight Summary

- ◆ Month-to-month contracts are the single strongest churn predictor
- ◆ Electronic check payment correlates with 33.6% of churned customers
- ◆ Low tenure (< 12 months) customers account for the majority of churn cases

- ◆ Absence of TechSupport doubles the likelihood of service migration
- ◆ High monthly charges (> \$60) are concentrated in the churned customer segment
- ◆ Gender shows no statistically significant impact on churn behavior

SECTION 05

Data Preprocessing Pipeline

5.1 Column Removal

The customerID column was removed as it carries no predictive value.

5.2 Data Type Correction

TotalCharges was stored as object type and was converted to numeric using coercion to handle blank string values.

5.3 Missing Value Handling

11 rows with tenure = 0 were identified and removed, ensuring dataset consistency. Final size: 7,032 records.

5.4 Categorical Encoding

All categorical features were encoded using LabelEncoder. Encoders were serialized as encoders.pkl for consistent deployment-time transformation.

5.5 Feature Scaling

Numerical features (tenure, MonthlyCharges, TotalCharges) were standardized using StandardScaler and saved as scaler.pkl.

5.6 Train-Test Split

A 70/30 stratified split was used to maintain class distribution, critical given the inherent class imbalance in churn datasets.

5.7 Class Imbalance Handling

Class imbalance was addressed within XGBoost using the scale_pos_weight parameter, improving recall for the minority churn class.

Feature Scaling Distribution

Standardizing numeric attributes

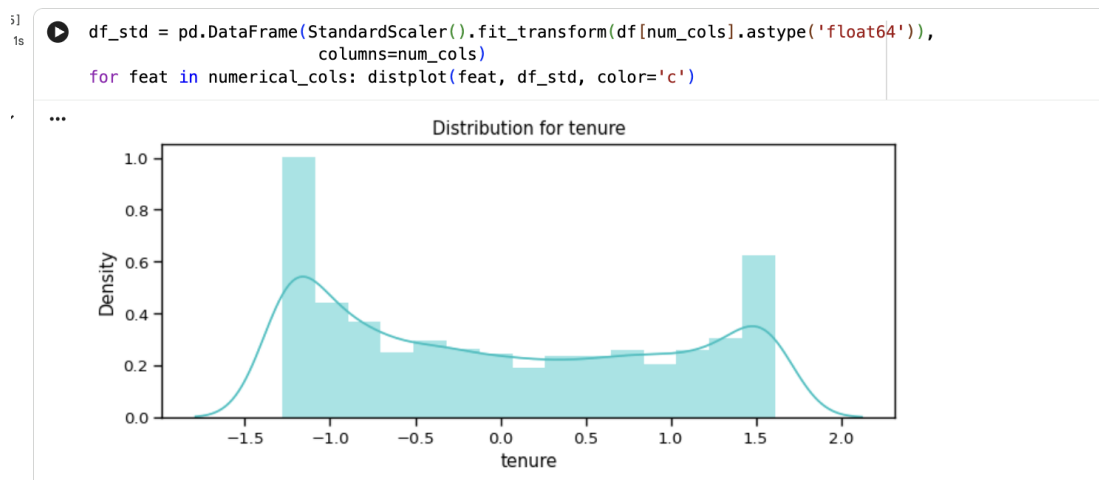


Figure 12: Standardized distribution of 'tenure' after applying StandardScaler

SECTION 06

Model Training & Evaluation

Three classification models were trained, evaluated, and compared. Given the business objective of **minimizing missed churn cases** (false negatives), **Recall** was prioritized as the primary metric.

Model	Accuracy	Recall (Churn)	Precision	F1-Score
Logistic Regression	80%	56%	64%	0.60
Decision Tree	78%	38%	64%	0.48
XGBoost (0.5 threshold)	75%	75%	52%	0.62
XGBoost (0.4 threshold) ✓	72%	81%	48%	0.60

Confusion Matrices

Confusion matrices provide a detailed breakdown of true positives, false positives, true negatives, and false negatives for each model.

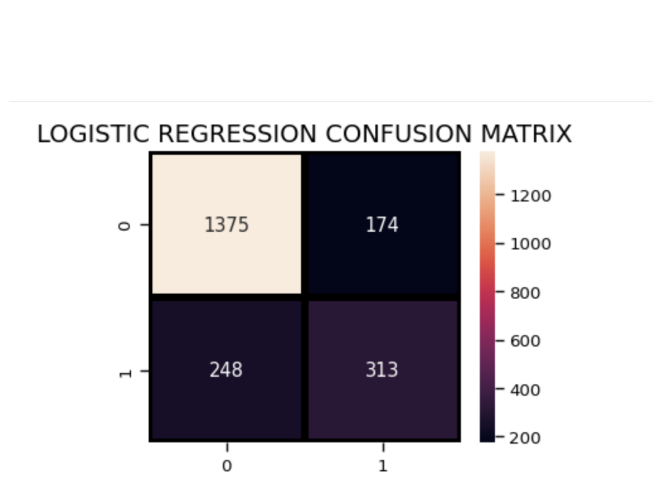


Figure 13: Logistic Regression Confusion Matrix

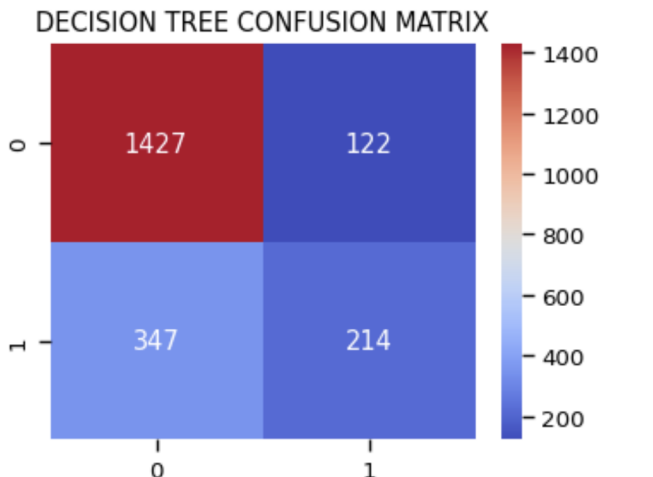


Figure 14: Decision Tree Confusion Matrix

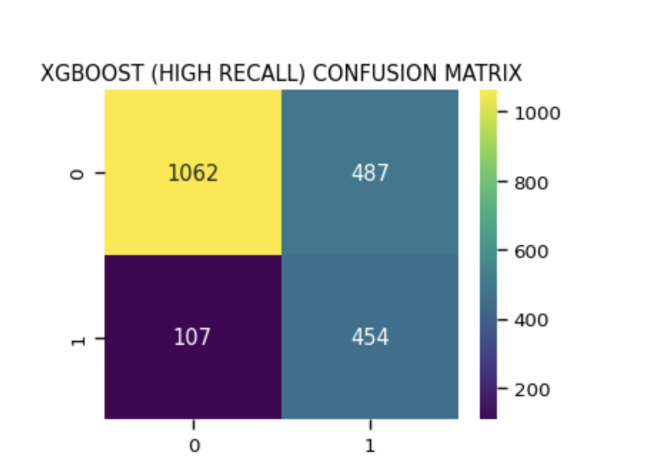
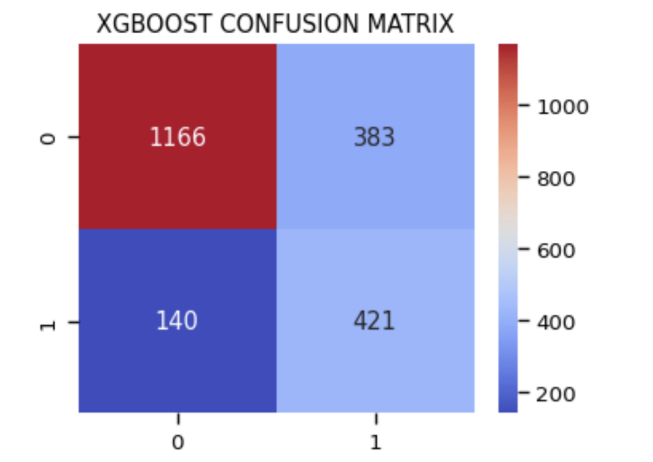


Figure 15: XGBoost (threshold=0.5) Confusion Matrix

Figure 16: XGBoost High-Recall (threshold=0.4) Confusion Matrix
— **SELECTED**

Model Analysis

Logistic Regression

Highest accuracy (80%) but recall of only 56% — misses nearly half of all actual churn customers, a significant business risk.

Decision Tree

Accuracy 78% but recall just 38% — weakest churn detection capability of all models evaluated.

XGBoost (0.5)

Accuracy 75%, recall 75% — strong churn detection at the default threshold.

XGBoost (0.4) ✓ **SELECTED**

Recall improved to 81% — best performance. Selected as the final model due to business-aligned optimization.

SECTION 07

Threshold Optimization

Instead of relying on the default binary classification, a probability-based threshold approach was implemented.

```
probabilities = model.predict_proba(X_test)[: , 1]
prediction = (probabilities > 0.4).astype(int)
```

This threshold reduction from 0.5 → 0.4 improved recall from 75% to **81%**. The threshold is serialized as **threshold.pkl** for consistency across environments.

Saved Model Artifacts

- final_churn_model.pkl
- scaler.pkl
- encoders.pkl
- threshold.pkl
- feature_order.pkl

SECTION 08

Streamlit Deployment

A production-ready Streamlit application was developed as the primary user interface. It features an interactive customer profile form with real-time churn probability scoring.

Application Screenshots

ML-POWERED ANALYTICS

Customer Churn Intelligence System

Predict customer retention behavior using machine learning — fast, accurate & actionable.

CUSTOMER PROFILE INPUT

GENDER

Male

SENIOR CITIZEN

No

PARTNER

No

DEPENDENTS

No

TENURE (MONTHS)

01272

PHONE SERVICE

No

MULTIPLE LINES

No

INTERNET SERVICE

DSL

ONLINE SECURITY

No

ONLINE BACKUP

No

DEVICE PROTECTION

No

TECH SUPPORT

No

STREAMING TV

No

STREAMING MOVIES

No

CONTRACT

Month-to-month

Figure 17: Customer Churn Intelligence System — Customer Profile Input Form

PAPERLESS BILLING

No

PAYMENT METHOD

Electronic check

MONTHLY CHARGES (\$)

70.00

TOTAL CHARGES (\$)

1500.00

Run Churn Prediction

Model Confidence Score: 0.41

PREDICTION RESULT

Customer Likely to Stay

Low churn risk — customer shows strong loyalty signals.

CHURN PROBABILITY SCORE

Threshold: 0.40

29.6%

Figure 18: Prediction result showing churn probability score and retention message

Application Features

- Interactive customer input form with all 19 features
- Real-time churn probability computation via XGBoost
- Custom threshold classification with configurable sensitivity
- Probability score visualization with gradient progress bar

- Retention messaging based on prediction outcome
- Model confidence score display for interpretability

Backend Inference Pipeline

Step	Component	Description
1	User Input	Customer profile collected via Streamlit form
2	Encoding	Categorical features transformed using saved LabelEncoders
3	Feature Order	Columns reordered to match training feature set exactly
4	Scaling	Numerical features normalized using saved StandardScaler
5	XGBoost Model	Trained model generates churn probability score
6	Threshold Logic	Probability compared against tuned 0.4 threshold
7	UI Output	Result displayed with prediction card and score bar

SECTION 09

Technology Stack

Component	Technology
Data Analysis	Pandas, NumPy
Visualization	Seaborn, Matplotlib, Plotly, Missingno
ML Models	Scikit-learn, XGBoost
Preprocessing	LabelEncoder, StandardScaler
Deployment	Streamlit (Cloud-hosted)
Model Storage	Joblib (.pkl artifacts)
Agent Framework (Planned)	LangGraph
Vector Store (Planned)	Chroma / FAISS
LLM (Planned)	Open-source (Free Tier)
Hosting (Planned)	Hugging Face / Streamlit Cloud / Render

SECTION 10

Team Contributions

The project was executed through a structured division of responsibilities ensuring clarity, efficiency, and accountability.

Vanshika Yadav

Data Analysis & Model Development Lead

- Interpreted the churn prediction problem from a business analytics perspective with focus on recall optimization
- Conducted deep dataset inspection, identified indirect missing values in TotalCharges column
- Implemented data type correction, categorical encoding (LabelEncoder), and StandardScaler normalization
- Trained and compared Logistic Regression, Decision Tree, and XGBoost models
- Performed threshold tuning from 0.5 → 0.4 to improve recall by 6 percentage points
- Serialized all model artifacts: final_churn_model.pkl, scaler.pkl, encoders.pkl, feature_order.pkl, threshold.pkl

Ronit Singh

Code Management, Technical Documentation & Engineering Workflow Lead

- Structured the Streamlit application backend using streamlit, pandas, numpy, and joblib
- Designed modular loading functions for all .pkl artifacts with @st.cache_resource optimization
- Ensured exact feature order alignment between training pipeline and production deployment
- Implemented probability-based classification with adjustable threshold logic (0.4) for recall optimization
- Added error handling and maintained clean project repository structure and README documentation
- Managed version control, synchronization between training notebook and production application, and technical documentation

Riya Garg

EDA, Visualization, Model Interpretation & Deployment Lead

- Conducted detailed EDA across all categorical and numerical features identifying key churn patterns
- Created comprehensive visualizations using Seaborn, Matplotlib, and Plotly (histograms, KDE plots, box plots, pie charts)
- Interpreted classification reports, confusion matrices, and F1-scores from a business perspective
- Identified and documented key churn drivers with actionable business recommendations
- Assisted in aligning visual insights with the Streamlit UI and verified probability outputs

Sankalp

Testing, Validation & GUI Development Lead

- Designed and implemented the interactive customer input form using st.columns() for organized data entry
- Developed visual components including prediction result cards, probability score display, and progress bars
- Tested end-to-end prediction pipeline from input encoding through model inference to threshold classification
- Conducted validation using model_test.py and confirmed alignment between notebook and deployment outputs
- Verified confusion matrices and confirmed threshold-based recall improvement across multiple test cases
- Ensured system stability, smooth UI interactions, and responsive interface design

SECTION 11

Contribution Matrix & Deliverables

Team Member	Dataset & Sourcing	Data Cleaning	EDA & Viz	Model Dev	Deployment	Docs
Vanshika Yadav	✓	✓	—	✓	—	✓
Ronit Singh	✓	✓	—	✓	—	✓
Riya Garg	✓	✓	✓	—	✓	—
Sankalp	—	✓	—	✓	✓	✓

Milestone 1 Deliverables — Completed

- Business understanding and problem framing
- Data cleaning and preprocessing pipeline
- Exploratory Data Analysis and visualization suite
- Multi-model comparison (Logistic Regression, Decision Tree, XGBoost)
- Threshold optimization for business-aligned recall improvement
- Churn probability generation and confidence scoring
- Model persistence with joblib artifacts
- Production Streamlit deployment (live application)
- Performance evaluation with confusion matrices and classification reports

Milestone 2 — Agentic Extension (Planned)

- LangGraph-based agent orchestration for autonomous retention strategy planning
- RAG pipeline using Chroma/FAISS for retrieval-augmented business recommendations
- Automated retention report generation using open-source LLMs
- Cloud deployment via Hugging Face Spaces or Render

SECTION 12

Future Improvements

- SHAP-based explainability for individual churn prediction transparency
- Feature importance dashboard for business stakeholder interpretation
- CRM system integration for automated retention workflow triggering
- Dynamic threshold tuning based on business cost-benefit analysis
- Customer segmentation module for targeted retention strategy personalization
- Agent-based autonomous retention planner with multi-step reasoning

SECTION 13

Conclusion

This project successfully implements a **complete end-to-end churn prediction system** — from raw data ingestion to a publicly deployed, production-ready application. The system demonstrates structured ML pipeline design, business-driven optimization, recall-based model selection, and full deployment readiness.

The final XGBoost model, tuned at a 0.4 classification threshold, correctly identifies **81% of at-risk customers** — providing the telecom business with an actionable early-warning system that can substantially reduce revenue loss from customer attrition.

With a clear path to Milestone 2, the system is architected to evolve from a predictive analytics tool into an intelligent, autonomous retention strategist powered by agentic AI.

- ◆ 81% recall — detects 4 out of 5 churning customers before they leave
- ◆ Full ML pipeline: EDA → Preprocessing → Training → Threshold Tuning → Deployment
- ◆ Live application: customerchurnprediction-2k327gcbblu4dawrhawsit.streamlit.app
- ◆ Open-source codebase: github.com/connectwithvanshika/customer_churn_prediction

Live Links & Resources

Resource	URL
Live Application	https://customerchurnprediction-2k327gcbblu4dawrhawsit.streamlit.app/
GitHub Repository	https://github.com/connectwithvanshika/customer_churn_prediction
Dataset (Kaggle)	https://www.kaggle.com/datasets/blastchar/telco-customer-churn
Google Colab Notebook	https://colab.research.google.com/drive/1qUUYKSU4QDwKlGH_H9j1Cr2NyEKXcqIA

— End of Report —

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